

Systematic Search for Singularities in 3D Euler-Voigt Flows

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Agenda

Introduction

- Singularities & Euler-Voigt Flows
- Finite-time Blow-up Criteria
- Initial Conditions

Optimization Problem for Euler-Voigt

- Formulation of the Optimization Problem
- Evaluation of the Gradient
- Riemannian Conjugate Gradient Method

Results

Singularities & Euler-Voigt Flows

- Navier-Stokes system ($\Omega = [0, L]^3$)

$$\begin{cases} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{0}, & \text{in } \Omega \times (0, T] \\ \nabla \cdot \mathbf{u} = 0, & \text{in } \Omega \times (0, T] \\ \mathbf{u} = \boldsymbol{\eta} & \text{in } \Omega \text{ at } t = 0 \\ \text{Periodic Boundary Conditions} & \text{on } \partial\Omega \times (0, T] \end{cases}$$

- The Big Question:

Given a smooth initial condition $\boldsymbol{\eta}$, does the Navier-Stokes system always admit smooth solutions $\mathbf{u}(t)$ for arbitrarily long times t ?

- One of the Clay Institute “Millennium Problems” (\$ 1M prize!)

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- $\alpha > 0$ is the regularization parameter with units of length. Acts as a 'spectral filter'. The Euler-Voigt system is globally well-posed given smooth initial conditions (Y. Cao et al. 2006).
- As $\alpha \rightarrow 0$ the solutions of Euler-Voigt converge to the solutions of Euler and so its behaviour can be used to determine if the Euler equations will develop a finite-time singularity.

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- Proven by Theorem 1 in (A. Larios et al. 2018).

Theorem

Let $\eta_\alpha \in H^s$, $s \geq 3$ with $\nabla \cdot \eta_\alpha = 0$ and let $\mathbf{u}_\alpha$ be the corresponding unique solution of Euler-Voigt. Suppose that there exists a time $T^* > 0$ such that

$$\limsup_{\alpha \rightarrow 0^+} \left(\alpha \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \right) > 0. \quad (1)$$

Then the Euler equations with the initial condition η_α must develop a singularity within the interval $[0, T^*]$.

- Theorem 5.1 (A. Larios et al. 2010).

Theorem

Given initial data $\eta, \eta_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$, let

$$\mathbf{u}, \mathbf{u}_\alpha \in C([0, T], H^s(\mathbb{T}^3)) \cap C^1([0, T], H^{s-1}(\mathbb{T}^3))$$

be the corresponding solutions to Euler and Euler-Voigt, respectively, where $0 < T < T_{\max}$ and $T_{\max}$ is the maximal time for which a solution to Euler exists and is unique. For all $t \in [0, T]$, we have

$$\begin{aligned} \|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \leq \\ (\|\eta - \eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\eta - \eta_\alpha)\|_{L^2}^2) e^{Ct} + C\alpha^2(e^{Ct} - 1). \end{aligned}$$

- Modified from Theorem 5.2 (A. Larios et al. 2010).

Theorem

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\Omega)$ for some $s \geq 3$ satisfying

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0,$$

let $\mathbf{u}, \mathbf{u}_\alpha \in C([0, T], H^s(\Omega)) \cap C^1([0, T], H^{s-1}(\Omega))$ be the corresponding solutions to Euler and Euler-Voigt, respectively. If there exists a finite time $T^* > 0$ such that the solutions $\mathbf{u}_\alpha$ satisfies

$$\limsup_{\alpha \rightarrow 0} \left(\alpha^2 \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) > 0,$$

then $\mathbf{u}$ develops a singularity in the interval $[0, T^*]$.

Proof.

Suppose that the solution $\mathbf{u}$ of the Euler equations Euler with the initial condition $\boldsymbol{\eta}$ is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have

$$\|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}^2,$$

thus

$$\alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}^2 - \|\mathbf{u}_\alpha(t)\|_{L^2}^2.$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2}^2.$$

Proof (Cont.)

Taking the limsup as $\alpha \rightarrow 0$, we obtain that

$$\begin{aligned} \limsup_{\alpha \rightarrow 0} \left(\sup_{t \in [0, T^*]} \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 \right) &= \limsup_{\alpha \rightarrow 0} \left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2}^2 \right) \\ &= \|\boldsymbol{\eta}\|_{L^2}^2 - \|b\mathbf{e}\|_{L^2}^2 \\ &= 0 \end{aligned}$$

which contradicts the hypothesis in the theorem that the above expression is positive.

Proof (Cont.)

The second equality holds because

$$\begin{aligned}
 \limsup_{\alpha \rightarrow 0} \left| \inf_{t \in [0, T^*]} \|\mathbf{u}_\alpha(t)\|_{L^2} - \|\boldsymbol{\eta}\|_{L^2} \right| &\leq \limsup_{\alpha \rightarrow 0} \|\mathbf{u}_\alpha(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\
 &\leq \limsup_{\alpha \rightarrow 0} \|\mathbf{u}_\alpha(t_\alpha) - \mathbf{u}(t_\alpha)\|_{L^2}^2 + \|\mathbf{u}(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\
 &= 0.
 \end{aligned}$$

Here we use the fact that the convergence of the solutions of the Euler-Voigt equations to the solution of the Euler equations is uniform on $[0, T^*]$, as shown in Theorem 5.1. □

- ▶ We consider analytic initial conditions that belong to the following Gevrey class

$$G^\sigma := \left\{ \mathbf{v} \in H^s(\Omega) : \|\mathbf{v}\|_{G^\sigma} = \langle \mathbf{v}, \mathbf{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0.$$

- ▶ We select the following inner product

$$\begin{aligned} \forall \mathbf{v}, \mathbf{u} \in G^\sigma, \quad \langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} &:= \sum_{\mathbf{j} \in \mathbb{Z}^3} (1 + |2\pi \mathbf{j}|^2)^s e^{4\pi\sigma|\mathbf{j}|} \widehat{\mathbf{v}}_{\mathbf{j}} \cdot \overline{\widehat{\mathbf{u}}_{\mathbf{j}}} \\ &= \int_{\Omega} (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v} \cdot \mathbf{u} \, d\mathbf{x}. \\ &= \left\langle (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v}, \mathbf{u} \right\rangle_{L^2} \end{aligned}$$

- ▶ The operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D| \mathbf{v}} \right]_{\mathbf{j}} := 2\pi |\mathbf{j}| \widehat{\mathbf{v}}_{\mathbf{j}}, \quad \left[\widehat{e^{\sigma|D|} \mathbf{v}} \right]_{\mathbf{j}} := e^{2\pi\sigma|\mathbf{j}|} \widehat{\mathbf{v}}_{\mathbf{j}}.$$

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- We assume zero mean velocity without loss of generality.

$$V := \{\mathbf{v} \in G^\sigma : \nabla \cdot \mathbf{v} = 0, \int_{\Omega} \mathbf{v} \, d\mathbf{x} = 0\}.$$

- We use a closed manifold $\mathcal{M}_1 \subset V$ defined as

$$\mathcal{M}_1 = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = \mathbf{C}\}.$$

- Euler ($\mathbf{u}$) and Euler-Voigt ($\mathbf{u}_\alpha$) time scaling property

$$\tilde{\mathbf{u}}_{(\alpha)}(\mathbf{x}, t) = \lambda \mathbf{u}_{(\alpha)}(\mathbf{x}, \lambda t), \quad (6)$$

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- Taylor Green Vortex ($\Omega = [0, L]^3$)

$$\eta_{TGV} = \begin{bmatrix} V_0 \sin(2\pi x/L) \cos(2\pi y/L) \cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L) \sin(2\pi y/L) \cos(2\pi z/L) \\ 0 \end{bmatrix}$$

- Characteristic time-scale

$$t_c = \frac{L}{V_0}$$

- Using ($\Omega = [0, 2\pi]^3$) and $V_0 = 1$, Bustamente et al. (2012) showed numerical evidence for finite-time singularity in Euler from η_{TGV} at $T^* \approx 4$. We use ($\Omega = [0, 1]^3$) and so scale down to $V_0 = 1/2\pi$.

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- Calculate C analytically

$$C = \|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla \boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx \right]^{1/2}$$

- Calculating for scaled $\boldsymbol{\eta}_{TGV}$

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \sqrt{3}/2$$

- Create the manifold

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- Beale-Kato-Majda (BKM) criterion (Beale et al. 1984): A Smooth solution u of the Euler system develops a singularity at $t = t_0$ if and only if

$$\lim_{t \rightarrow t_0} \int_0^t \|\omega\|_{L^\infty} d\tau = \infty, \quad \omega = \nabla \times u$$

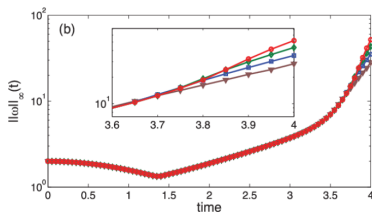


Figure: (Bustamente et al. 2012 Fig. 1b) Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 512^3 (brown triangles), 1024^3 (blue squares), 2048^3 (green diamonds), and 4096^3 (red circles).

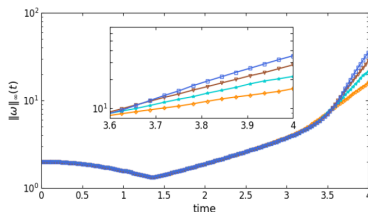


Figure: Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), 1024^3 (blue squares).

Optimization Problem for Euler-Voigt

- Define our objective function $\Phi_{\alpha,T} : V \rightarrow \mathbb{R}^+$

$$\Phi_{\alpha,T}(\eta) := \|\nabla \mathbf{u}_\alpha(T; \eta_\alpha)\|_{L^2}^2 = \|\mathbf{u}_\alpha(T; \eta_\alpha)\|_{\dot{H}^1}^2$$

- Given $\alpha, T \in \mathbb{R}^+$, find

$$\tilde{\eta}_{\alpha,T} = \operatorname{argmax}_{\eta_\alpha \in \mathcal{M}_1} \Phi_{\alpha,T}(\eta_\alpha).$$

- If $\alpha^2 \Phi_{\alpha,T}(\tilde{\eta}_{\alpha,T}) > 0$ as $\alpha \rightarrow 0$ then a singularity forms in Euler the interval $[0, T]$.

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- ▶ The first order finite difference equation for the Gâteaux differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) : V \times V \rightarrow \mathbb{R}$,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)] .$$

- ▶ Substituting perturbations into Euler-Voigt

$$\boldsymbol{u}_\alpha \rightarrow \boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha, \boldsymbol{\eta}_\alpha \rightarrow \boldsymbol{\eta}_\alpha + \boldsymbol{\eta}'_\alpha, \text{ and } p \rightarrow p + p'$$

- ▶ Linearization of Euler-Voigt

$$\mathcal{L} \begin{bmatrix} \boldsymbol{u}'_\alpha \\ p' \end{bmatrix} := \begin{bmatrix} \partial_t \boldsymbol{u}'_\alpha + (I - \alpha^2 \Delta)^{-1} (\boldsymbol{u}_\alpha \cdot \nabla \boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha \cdot \nabla \boldsymbol{u}_\alpha + \nabla p') = 0 \\ \nabla \cdot \boldsymbol{u}'_\alpha \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boldsymbol{u}'_\alpha(x, 0) = \boldsymbol{\eta}'_\alpha$$

- ▶ The first order finite difference equation for the Gâteaux differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) : V \times V \rightarrow \mathbb{R}$,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)] .$$

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- Integration by parts on the following duality-pairing relation

$$\begin{aligned}
 \left(\mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p' \end{bmatrix}, \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^* \end{bmatrix} \right) &:= \int_0^T \int_{\mathbb{T}^3} \mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p' \end{bmatrix}, \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^* \end{bmatrix} d\mathbf{x} dt \\
 &= \left(\begin{bmatrix} \mathbf{u}'_\alpha \\ p' \end{bmatrix}, \mathcal{L}^* \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^* \end{bmatrix} \right) + \int_{\mathbb{T}^3} \mathbf{u}'_\alpha(\mathbf{x}, T) \cdot \mathbf{u}^*_\alpha(\mathbf{x}, T) d\mathbf{x} \\
 &\quad - \int_{\mathbb{T}^3} \boldsymbol{\eta}'(\mathbf{x}) \cdot \mathbf{u}^*_\alpha(\mathbf{x}, 0) d\mathbf{x} \\
 &= 0.
 \end{aligned}$$

- Adjoint System

$$\mathcal{L}^* \begin{bmatrix} \mathbf{u}^*_\alpha \\ p^*_\alpha \end{bmatrix} := \begin{bmatrix} -\partial_t \mathbf{u}^*_\alpha - (\nabla(I - \alpha^2)^{-1} \mathbf{u}^*_\alpha + (\nabla(I - \alpha^2)^{-1} \mathbf{u}^*_\alpha)^\top) \mathbf{u}_\alpha - \nabla p^*_\alpha \\ -\nabla \cdot \mathbf{u}^*_\alpha \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{u}^*_\alpha(\mathbf{x}, T) = 2|D|^2 \mathbf{u}_\alpha(\mathbf{x}, T)$$

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 &= \left(\begin{bmatrix} \mathbf{u}'_\alpha \\ \rho' \end{bmatrix}, \mathcal{L}^* \begin{bmatrix} \mathbf{u}^*_\alpha \\ \rho^* \end{bmatrix} \right) + \int_{\mathbb{T}^3} \mathbf{u}'_\alpha(\mathbf{x}, T) \cdot \mathbf{u}^*_\alpha(\mathbf{x}, T) d\mathbf{x} \\
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- ▶ Substituting $\Phi_{\alpha,T}$ into the Gâteaux differential,

$$\Phi'_{\alpha,T}(\eta; \eta') = 2 \langle \mathbf{u}_{\alpha}(\mathbf{x}, T), \mathbf{u}'_{\alpha}(\mathbf{x}, T) \rangle_{\dot{H}^1} = 2 \langle \mathbf{u}_{\alpha}^*(\mathbf{x}, 0), \mathbf{u}'_{\alpha}(\mathbf{x}, 0) \rangle_{L^2}^2$$

- ▶ From the Riesz representation theorem (Luenberger 1969)

$$\Phi'_{\alpha,T}(\eta : \eta') = \langle \nabla \Phi_{\alpha,T}(\eta_{\alpha}), \eta'_{\alpha} \rangle_{G^{\sigma}}$$

- ▶ The gradient of the objective functional is

$$\nabla \Phi_{\alpha,T}(\eta_{\alpha}) = (1 + |D|^2)^{-3} e^{-2\sigma|D|} \mathbf{u}_{\alpha}^*(x, 0)$$

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- ▶ Project the gradient $\nabla\Phi_T(\eta_\alpha)$ onto the tangent space to $\mathcal{M}_1$ at η_α .
- ▶ Perform vector transport operation to construct a Riemannian conjugate ascent direction.
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- Tangent space to $\mathcal{M}_1$ at $\mathbf{v}$

$$\mathcal{T}_{\mathbf{v}}\mathcal{M}_1 = \{z \in V : \langle \mathbf{v}, z \rangle_{\dot{H}^1} = 0\}$$

- For any $z \in \mathcal{T}_{\mathbf{v}}\mathcal{M}_1$

$$\begin{aligned} \langle \mathbf{v}, z \rangle_{\dot{H}^1} &= \int_{\mathbb{T}^3} |D|\mathbf{v} \cdot |D|z \, d\mathbf{x} \\ &= \int_{\mathbb{T}^3} (1 + |D|^2)^3 e^{2\sigma|D|} \left((1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v} \right) \cdot z \, d\mathbf{x} \\ &= \left\langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}, z \right\rangle_{G^\sigma} = 0 \end{aligned}$$

- Characterized using the unit 'normal' vector

$$\mathbf{n}_{\mathbf{v}} = \frac{(1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}}{\|(1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}\|_{G^\sigma}}$$

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- Projection operator $P_{\mathcal{T}_v\mathcal{M}_1} : V \rightarrow \mathcal{T}_v\mathcal{M}_1$

$$\forall \mathbf{w} \in V, \quad P_{\mathcal{T}_v\mathcal{M}_1} \mathbf{w} = \mathbf{w} - \langle \mathbf{n}_v, \mathbf{w} \rangle_{G^\sigma} \mathbf{n}_v$$

- Retraction operator $R : V \rightarrow \mathcal{M}_1$

$$R(\mathbf{v}) = \frac{\sqrt{3}}{2} \frac{\mathbf{v}}{\|\mathbf{v}\|_{\dot{H}^1}}, \quad \mathbf{v} \neq 0$$

- Vector field ξ transported along $\mathcal{M}_1$ in the direction of ϕ

$$\mathcal{TM}_1 \oplus \mathcal{TM}_1 \rightarrow \mathcal{TM}_1 : (\varphi, \xi) \mapsto \Gamma_\varphi(\xi) \in \mathcal{TM}_1,$$

where $\mathcal{TM}_1 = \cup_{\mathbf{v} \in \mathcal{M}_1} \mathcal{T}_v\mathcal{M}_1$ is the tangent bundle on $\mathcal{M}_1$ (absil et al. 2008). It is calculated via the differentiated retraction

$$\Gamma_\varphi(\xi) = \left. \frac{d}{dt} R(\varphi + t\xi) \right|_{t=0}.$$

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- We can now construct the iterative relation

$$\boldsymbol{\eta}_\alpha^{(n+1)} = R \left[\boldsymbol{\eta}_\alpha^{(n)} + \tau_n \mathbf{d}^{(n)} \right]$$

- Let $\mathbf{G}^{(n)} := \nabla \Phi_{\alpha, T}(\boldsymbol{\eta}_\alpha^{(n)})$. Let $\mathcal{T}_n := \mathcal{T}_{\boldsymbol{\eta}_\alpha^{(n)}} \mathcal{M}_1$. $\mathbf{d}^{(n)}$ is computed as

$$\mathbf{d}^{(0)} = P_{\mathcal{T}_0} \mathbf{G}^{(0)}$$

$$\mathbf{d}^{(n)} = P_{\mathcal{T}_n} \mathbf{G}^{(n)} + \beta_n \Gamma_{\tau_{n-1} \mathbf{d}^{(n-1)}}(\mathbf{d}^{(n-1)}), \quad n \geq 1.$$

- 'Momentum' term is computed using the Polak-Ribière approach (Nocedal et al. 1999).

$$\beta_n = \frac{\left\langle P_{\mathcal{T}_n} \mathbf{G}^{(n)}, \left(P_{\mathcal{T}_n} \mathbf{G}^{(n)} - \Gamma_{\tau_{n-1} \mathbf{d}^{(n-1)}} P_{\mathcal{T}_{n-1}} \mathbf{G}^{(n-1)} \right) \right\rangle_{G^\sigma}}{\|P_{\mathcal{T}_{n-1}} \mathbf{G}^{(n-1)}\|_{G^\sigma}^2}$$

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$$\tau_n = \operatorname{argmax}_{\tau > 0} \Phi_T \left(R \left[\boldsymbol{\eta}^{(n)} + \tau \mathbf{d}^{(n)} \right] \right)$$

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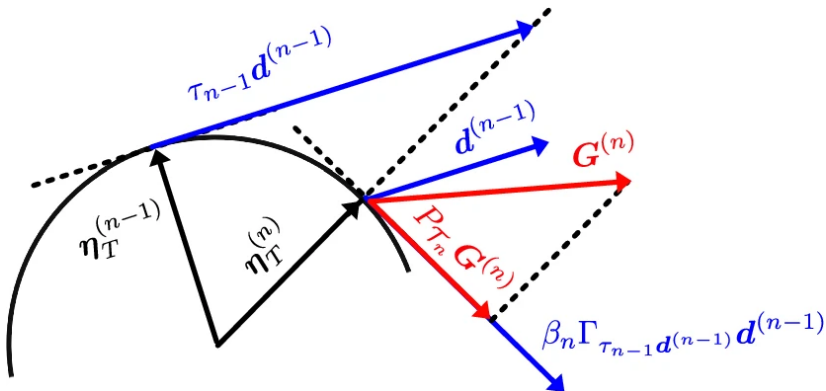


Figure: (Zhao et al. 2023, Fig. 1)

► Periodic algorithm restart ($\beta_n = 0$)



$$n = 20k, \quad k \in \mathbb{Z}^+$$

- The search direction $\mathbf{d}^{(n)}$ is not an ascent direction

$$\frac{\langle \mathbf{d}^{(n)}, P_{\mathcal{T}_n} \mathbf{G}^{(n)} \rangle_{G^\sigma}}{\|\mathbf{d}^{(n)}\|_{G^\sigma} \|P_{\mathcal{T}_n} \mathbf{G}^{(n)}\|_{G^\sigma}} < \epsilon, \quad 0 < \epsilon \ll 1.$$

- Iterations are stopped when the relative change of $\Phi_{\alpha, T}$ is below a specified tolerance

$$0 \leq \frac{\Phi_{\alpha, T}(\eta_\alpha^{(n+1)}) - \Phi_{\alpha, T}(\eta_\alpha^{(n)})}{\Phi_{\alpha, T}(\eta_\alpha^{(n)})} < tol.$$

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Numerical Methods

- ▶ Discretized in space using a standard pseudospectral Fourier method
- ▶ Derivatives are evaluated in the Fourier space
- ▶ Nonlinear products are computed in the physical space on an uniform grid with N points in each direction.
- ▶ Explicit fourth-order Runge-Kutta
- ▶ 2/3 dealiasing

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► Domain

resolution : $128^3, 256^3, 512^3, 1024^3$

$\alpha : 1/1024, 2/1024, 4/1024, 8/1024, 16/1024$

► Time

$dt : 2^{-5}$

$T : 3, 5$

► Gevrey class

$\sigma : 1E-5, 1E-4, 1E-3, 1E-2, 1E-1$

$s : 3$

► Domain

resolution : $128^3, 256^3, 512^3, 1024^3$

$\alpha : 1/1024, 2/1024, 4/1024, 8/1024, 16/1024$

► Time

$dt : 2^{-5}$

$T : 3, 5$

► Gevrey class

$\sigma : 1E-5, 1E-4, 1E-3, 1E-2, 1E-1$

$s : 3$

► Domain

resolution : $128^3, 256^3, 512^3, 1024^3$

$\alpha : 1/1024, 2/1024, 4/1024, 8/1024, 16/1024$

► Time

$dt : 2^{-5}$

$T : 3, 5$

► Gevrey class

$\sigma : 1\text{E-}5, 1\text{E-}4, 1\text{E-}3, 1\text{E-}2, 1\text{E-}1$

$s : 3$

Results

- Evolution of $\|\mathbf{u}_\alpha(\mathbf{x}, t)\|_{H^1}^2$ over time for $\alpha = 16/1024$.

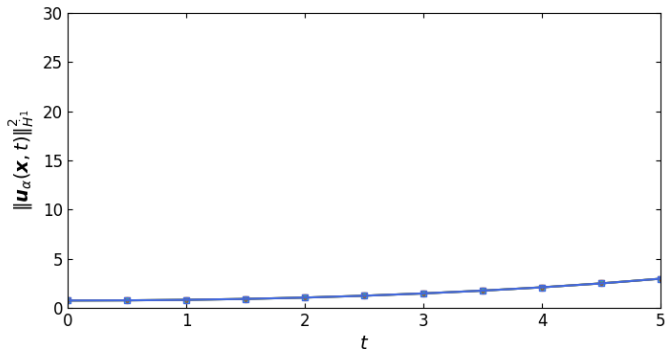


Figure: Results for $\alpha = 16/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\mathbf{u}_\alpha(\mathbf{x}, t)\|_{H^1}^2$ over time for $\alpha = 8/1024$.

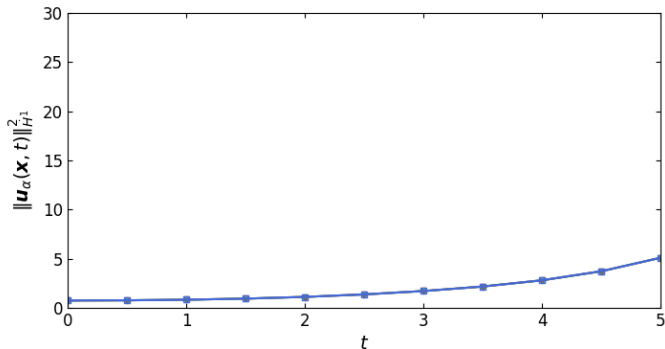


Figure: Results for $\alpha = 8/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\mathbf{u}_\alpha(\mathbf{x}, t)\|_{H^1}^2$ over time for $\alpha = 4/1024$.

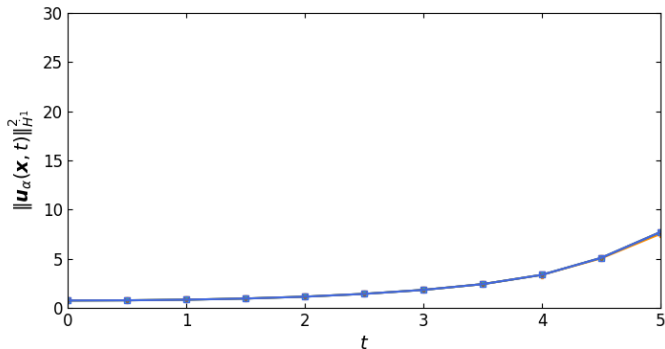


Figure: Results for $\alpha = 4/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\mathbf{u}_\alpha(\mathbf{x}, t)\|_{H^1}^2$ over time for $\alpha = 2/1024$.

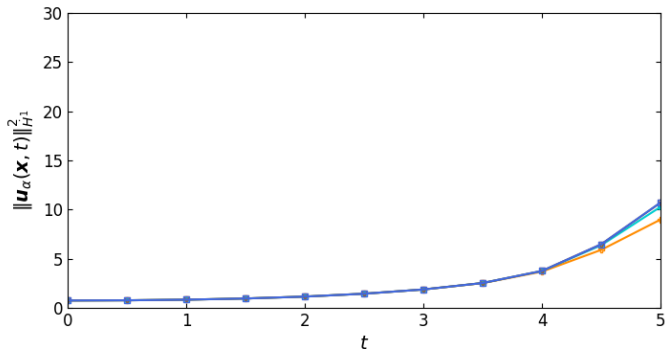


Figure: Results for $\alpha = 2/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\mathbf{u}_\alpha(\mathbf{x}, t)\|_{H^1}^2$ over time for $\alpha = 1/1024$.

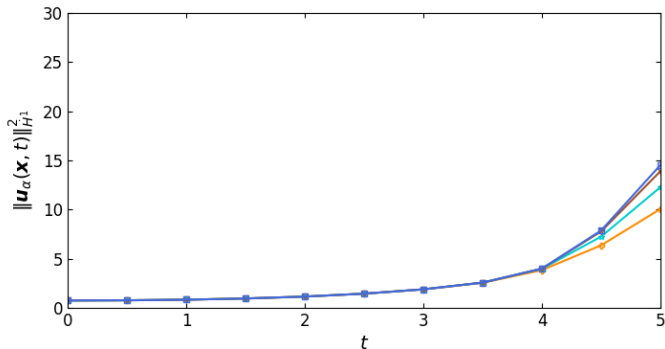


Figure: Results for $\alpha = 1/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\mathbf{u}(\mathbf{x}, t)\|_{\dot{H}^1}^2$ over time for $\alpha = 0$.

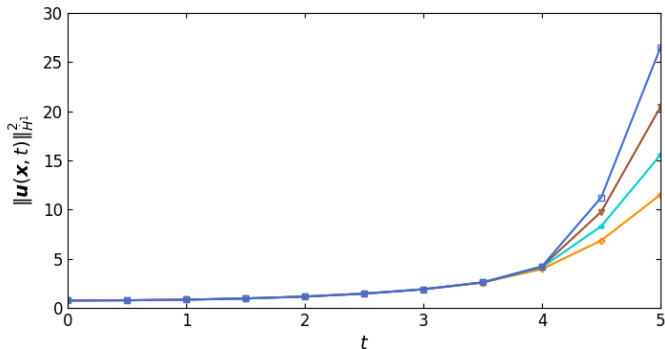


Figure: Results for $\alpha = 0$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Results of $\alpha^2 \Phi_{\alpha, T}(\eta_{\alpha}^0)$ as $\alpha \rightarrow 0$ for each resolution at $T = 3, 5$.

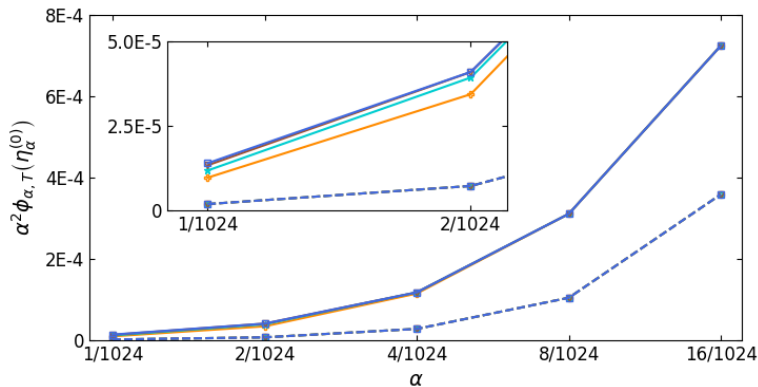


Figure: Results for $T = 3$ (dotted line) and for $T = 5$ (solid line). Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).